RIG-SURJECTIVITY VERSUS FAITHFUL FLATNESS IN DEGREE ONE UNDER VANISHING I-TORSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete in both directions

Editorial note. The label lemma-flat-rig-surjective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- formal spectra and rig-surjectivity tested on adic maps to complete discrete valuation rings
- the faithfully-flat case of the preceding rig-surjectivity lemma
- vanishing of I-torsion versus I-power torsion and the Noetherian annihilator lemma
- the Jacobson lemma on primes defining closed points of the complement of a closed point
- embedding of a one-dimensional local domain into a discrete valuation ring and faithful flatness of maps of rings detected on spectra

Source reference. Original lemma 00L6: sources/stacks-chow/tags.txt, source lines 776–776. Prerequisite pointer invoked at proof line(s) 5497; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00PH: sources/stacks-chow/tags.txt, source lines 931–931. Prerequisite pointer invoked at proof line(s) 5506; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00HQ: sources/stacks-chow/tags.txt, source lines 652–652. Prerequisite pointer invoked at proof line(s) 5521; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AN0: sources/stacks-chow/tags.txt, source lines 13423–13423. Prerequisite pointer invoked at proof line(s) 5489; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AQX: sources/stacks-restricted/pinned-original-source.tex, source lines 5418–5418. Prerequisite pointer invoked at proof line(s)

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5482; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05EA: sources/stacks-chow/tags.txt, source lines 6983–6983. Prerequisite pointer invoked at proof line(s) 5485; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02IM: sources/stacks-chow/tags.txt, source lines 3305–3305. Prerequisite pointer invoked at proof line(s) 5502; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a Noetherian ring complete with respect to an ideal I. Let B be an I-adically complete A-algebra. Assume that*

(1) *the I-torsion in A is 0,*
(2) *$A/I^n \to B/I^nB$ is flat and of finite type for all n.*

Then Spf$(B) \to$ Spf(A) is rig-surjective if and only if $A/I \to B/IB$ is faithfully flat.

Proof. Faithful flatness implies rig-surjectivity by Lemma 0AQX. To prove the converse we will use without further mention that the vanishing of I-torsion is equivalent to the vanishing of I-power torsion (More on Algebra, Lemma 05EA). We will also use without further mention that morphisms between formal spectra are given by continuous maps between the corresponding topological rings, see Formal Spaces, Lemma 0AN0.

Assume $\mathrm{Spf}(B) \to \mathrm{Spf}(A)$ is rig-surjective. Choose a maximal ideal $I \subset \mathfrak{m} \subset A$. The open $U = \mathrm{Spec}(A_\mathfrak{m}) \setminus V(I_\mathfrak{m})$ of $\mathrm{Spec}(A_\mathfrak{m})$ is nonempty as the $I_\mathfrak{m}$-torsion of $A_\mathfrak{m}$ is zero (use Algebra, Lemma 00L6). Thus we can find a prime $\mathfrak{q} \subset A_\mathfrak{m}$ which defines a point of U (i.e., $IA_\mathfrak{m} \not\subset \mathfrak{q}$) and which corresponds to a closed point of $\mathrm{Spec}(A_\mathfrak{m}) \setminus \{\mathfrak{m}\}$, see Properties, Lemma 02IM. Then $A_\mathfrak{m}/\mathfrak{q}$ is a dimension 1 local domain. Thus we can find an injective local homomorphism of local rings $A_\mathfrak{m}/\mathfrak{q} \subset R$ where R is a discrete valuation ring (Algebra, Lemma 00PH). By construction $IR \subset \mathfrak{m}_R$ and we see that $A \to R$ extends to a continuous map $A \to R^\wedge$. Since the completion of a discrete valuation ring is a discrete valuation ring, we see that the assumption gives a commutative diagram of rings

$$\begin{array}{ccc} R' & \longleftarrow & B \\ \uparrow & & \uparrow \\ R^\wedge & \longleftarrow & A \end{array}$$

Thus we find a prime ideal of B lying over $\mathfrak{m}$. It follows that $\mathrm{Spec}(B/IB) \to \mathrm{Spec}(A/I)$ is surjective, whence $A/I \to B/IB$ is faithfully flat (Algebra, Lemma 00HQ). □